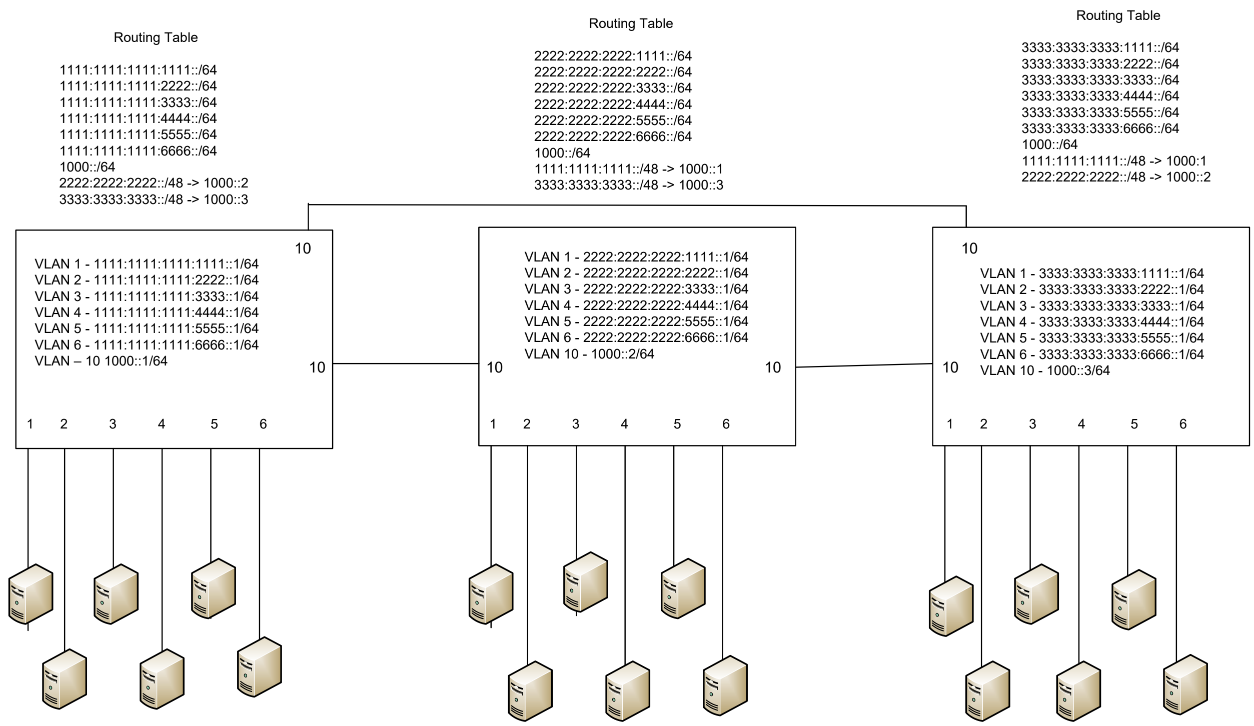
Lab 12

IPV6

Objective:

The purpose of this lab is for the student to understand, create and implement IPV6. Students will also utilize IOS commands to explore and configure IPV6 on network routers and switches.

Diagram:



Components:

18 – PC’s with an Ethernet NIC

3 – Layer 3 Switches

Procedure:

1. Connect all PC's together and put their static IP addresses, subnet masks and gateways on them.

2. Label the Layer 3 switches VLAN IP addresses below

L3 Switch 1

VLAN	Sw. Port	IP Address
1	1	1111:1111:1111:1111::1/64
2	2	1111:1111:1111:2222::1/64
3	3	1111:1111:1111:3333::1/64
4	4	1111:1111:1111:4444::1/64
5	5	1111:1111:1111:5555::1/64
6	6	1111:1111:1111:6666::1/64
10	10&11	1000::1/64

L3 Switch 2

VLAN	Sw. Port	IP Address
1	1	2222:2222:2222:1111::1/64
2	2	2222:2222:2222:2222::1/64
3	3	2222:2222:2222:3333::1/64
4	4	2222:2222:2222:4444::1/64
5	5	2222:2222:2222:5555::1/64
6	6	2222:2222:2222:6666::1/64
10	10&11	1000::2/64

L3 Switch 3

VLAN	Sw. Port	IP Address
1	1	3333:3333:3333:1111::1/64
2	2	3333:3333:3333:2222::1/64
3	3	3333:3333:3333:3333::1/64
4	4	3333:3333:3333:4444::1/64
5	5	3333:3333:3333:5555::1/64
6	6	3333:3333:3333:6666::1/64
10	10&11	1000::3/64

3. Label the routing tables for the Layer 3 switch and routers below.

L3 Switch 1	L3 Switch 2	L3 Switch 3
C-1111:1111:1111:1111::1/64	C-2222:2222:2222:1111::1/64	C-3333:3333:3333:1111::1/64
C-1111:1111:1111:2222::1/64	C-2222:2222:2222:2222::1/64	C-3333:3333:3333:2222::1/64
C-1111:1111:1111:3333::1/64	C-2222:2222:2222:3333::1/64	C-3333:3333:3333:3333::1/64
C-1111:1111:1111:4444::1/64	C-2222:2222:2222:4444::1/64	C-3333:3333:3333:4444::1/64
C-1111:1111:1111:5555::1/64	C-2222:2222:2222:5555::1/64	C-3333:3333:3333:5555::1/64
C-1111:1111:1111:6666::1/64	C-2222:2222:2222:6666::1/64	C-3333:3333:3333:6666::1/64
C-1000::1/64	C-1000::2/64	C-1000::3/64
S-2222:2222:2222::/48 -> 1000::2	S-1111:1111:1111::/48 -> 1000::1	S-1111:1111:1111::/48 -> 1000::1
S-3333:3333:3333::/48 -> 1000::3	S-3333:3333:3333::/48 -> 1000::3	S-2222:2222:2222::/48 -> 1000::2

4. Place the switch ports into the proper vlans. Configure all of the switches vlan IP addresses.

5. Enable ipv6 routing on the Layer 3 switches

6. On switch 1 enable router EIGRP as follows:
    a. Make the AS number 1
        i. What command did you enter
           Ipv6 router eigrp 1
    b. Make the EIGRP ID 10.10.10.1
        i. What command did you enter
           Eigrp router-id 10.10.10.1
    c. Turn on EIGRP
        i. What command did you enter
           No shut
    d. Enable EIGRP on the vlans
        i. What commands did you enter
           int vlan 1
           ipv6 eigrp 1

int vlan 2
ipv6 eigrp 1
int vlan 3
ipv6 eigrp 1
int vlan 4
ipv6 eigrp 1
int vlan 5
ipv6 eigrp 1
int vlan 6
ipv6 eigrp 1
int vlan 10
ipv6 eigrp 1

7. On switch 2 enable router EIGRP as follows:
   a. Make the AS number 1
      i. What command did you enter
         Ipv6 router eigrp 1
   b. Make the EIGRP ID 10.10.10.2
      i. What command did you enter
         Eigrp router-id 10.10.10.2
   c. Turn on EIGRP
      i. What command did you enter
         No shut
   d. Enable EIGRP on the vlans
      i. What commands did you enter
         int vlan 1
         ipv6 eigrp 1
         int vlan 2
         ipv6 eigrp 1
         int vlan 3
         ipv6 eigrp 1
         int vlan 4

ipv6 eigrp 1
int vlan 5
ipv6 eigrp 1
int vlan 6
ipv6 eigrp 1
int vlan 10
ipv6 eigrp 1

8. On switch 3 enable router EIGRP as follows:
    a. Make the AS number 1
        i. What command did you enter
        Ipv6 router eigrp 1
    b. Make the EIGRP ID 10.10.10.3
        i. What command did you enter
        Eigrp router-id 10.10.10.3
    c. Turn on EIGRP
        i. What command did you enter
        no shut
    d. Enable EIGRP on the vlans
        i. What commands did you enter
        int vlan 1
        ipv6 eigrp 1
        int vlan 2
        ipv6 eigrp 1
        int vlan 3
        ipv6 eigrp 1
        int vlan 4
        ipv6 eigrp 1
        int vlan 5
        ipv6 eigrp 1
        int vlan 6
        ipv6 eigrp 1

int vlan 10
ipv6 eigrp 1

9. Verify the routing table on the Layer 3 switches
    a. What command did you enter? show ipv6 route

10. Can the computers ping each other? no

11. Remove EIGRP from all 3 Layer 3 switches and all VLAN interfaces

12. On Switch 1 add static routes for 2222:2222:2222::/48 and 3333:3333:3333::/48
    a. What commands did you enter
       Ipv6 route 2222:2222:2222::/48 1000::2
       Ipv6 route 3333:3333:3333::/48 1000::3

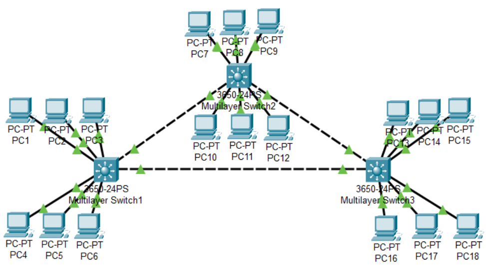
13. On Switch 2 add static routes for 1111:1111:1111::/48 and 3333:3333:3333::/48
    a. What commands did you enter
       Ipv6 route 1111:1111:1111::/48 1000::1
       Ipv6 route 3333:3333:3333::/48 1000::3

14. On Switch 3 add static routes for 1111:1111:1111::/48 and 2222:2222:2222::/48
    a. What commands did you enter
       Ipv6 route 1111:1111:1111::/48 1000::1
       Ipv6 route 2222:2222:2222::/48 1000::2

15. Verify the routing table on the Layer 3 switches
    a. What command did you enter? show ipv6 route

16. Can the computers ping each other? No, but they’re supposed to

17. Create the lab diagram below (PCs, router and switch connections) in Packet Tracer or Visio. **<u>Label the IP addresses, subnet masks, and gateways</u>** used by each device. **<u>Label the routing tables next to the switch and router</u>** and the **<u>switches vlan IP addresses and port vlan information</u>**.




```
PC 1 IP-1111:1111:1111:1111::10/64
Gateway 1111:1111:1111:1111::1
PC 2 IP-1111:1111:1111:2222::10/64
Gateway 1111:1111:1111:2222::1
PC 3 IP-1111:1111:1111:3333::10/64
Gateway 1111:1111:1111:3333::1
PC 4 IP-1111:1111:1111:4444::10/64
Gateway 1111:1111:1111:4444::1
PC 5 IP-1111:1111:1111:5555::10/64
Gateway 1111:1111:1111:5555::1
PC 6 IP-1111:1111:1111:6666::10/64
Gateway 1111:1111:1111:6666::1
```

```
PC 7 IP-2222:2222:2222:1111::10/64
Gateway 2222:2222:2222:1111::1
PC 8 IP-2222:2222:2222:2222::10/64
Gateway 2222:2222:2222:2222::1
PC 9 IP-2222:2222:2222:3333::10/64
Gateway 2222:2222:2222:3333::1
PC 10 IP-2222:2222:2222:4444::10/64
Gateway 2222:2222:2222:4444::1
PC 11 IP-2222:2222:2222:5555::10/64
Gateway 2222:2222:2222:5555::1
PC 12 IP-2222:2222:2222:6666::10/64
Gateway 2222:2222:2222:6666::1
```

```
PC 13 IP-3333:3333:3333:1111::10/64
Gateway 3333:3333:3333:1111::1
PC 14 IP-3333:3333:3333:2222::10/64
Gateway 3333:3333:3333:2222::1
PC 15 IP-3333:3333:3333:3333::10/64
Gateway 3333:3333:3333:3333::1
PC 16 IP-3333:3333:3333:4444::10/64
Gateway 3333:3333:3333:4444::1
PC 17 IP-3333:3333:3333:5555::10/64
Gateway 3333:3333:3333:5555::1
PC 18 IP-3333:3333:3333:6666::10/64
Gateway 3333:3333:3333:6666::1
```

```
L3 Switch 1 VLAN Table
VLAN Name                          Status    Ports
---- ----------------------------- --------- ----------------------------
1    default                       active    Po1, Gig1/0/1,
2    VLAN0002                      active    Po2, Gig1/0/2
3    VLAN0003                      active    Po3, Gig1/0/3
4    VLAN0004                      active    Po4, Gig1/0/4
5    VLAN0005                      active    Po5, Gig1/0/5
6    VLAN0006                      active    Po6, Gig1/0/6
10   VLAN0010                      active
Vlan1                  [up/up]
    FE80::201:43FF:FE84:C1ED
    1111:1111:1111:1111::1
Vlan2                  [up/up]
    FE80::201:43FF:FE84:C101
    1111:1111:1111:2222::1
Vlan3                  [up/up]
    FE80::201:43FF:FE84:C102
    1111:1111:1111:3333::1
Vlan4                  [up/up]
    FE80::201:43FF:FE84:C103
    1111:1111:1111:4444::1
Vlan5                  [up/up]
    FE80::201:43FF:FE84:C104
    1111:1111:1111:5555::1
Vlan6                  [up/up]
    FE80::201:43FF:FE84:C105
    1111:1111:1111:6666::1
Vlan10                 [up/up]
    FE80::201:43FF:FE84:C106
    1000::1
```

```
L3 Switch 2 VLAN Table
VLAN Name                          Status    Ports
---- ----------------------------- --------- ----------------------------
1    default                       active    Po1, Gig1/0/1,
2    VLAN0002                      active    Po2, Gig1/0/2
3    VLAN0003                      active    Po3, Gig1/0/3
4    VLAN0004                      active    Po4, Gig1/0/4
5    VLAN0005                      active    Po5, Gig1/0/5
6    VLAN0006                      active    Po6, Gig1/0/6
10   VLAN0010                      active
Vlan1                  [up/up]
    FE80::20A:F3FF:FE71:AA88
    2222:2222:2222:1111::1
Vlan2                  [up/up]
    FE80::20A:F3FF:FE71:AA01
    2222:2222:2222:2222::1
Vlan3                  [up/up]
    FE80::20A:F3FF:FE71:AA02
    2222:2222:2222:3333::1
Vlan4                  [up/up]
    FE80::20A:F3FF:FE71:AA03
    2222:2222:2222:4444::1
Vlan5                  [up/up]
    FE80::20A:F3FF:FE71:AA04
    2222:2222:2222:5555::1
Vlan6                  [up/up]
    FE80::20A:F3FF:FE71:AA05
    2222:2222:2222:6666::1
Vlan10                 [up/up]
    FE80::20A:F3FF:FE71:AA06
    1000::2
```

```
L3 Switch 3 VLAN Table
VLAN Name                          Status    Ports
---- ----------------------------- --------- ----------------------------
1    default                       active    Po1, Gig1/0/1,
2    VLAN0002                      active    Po2, Gig1/0/2
3    VLAN0003                      active    Po3, Gig1/0/3
4    VLAN0004                      active    Po4, Gig1/0/4
5    VLAN0005                      active    Po5, Gig1/0/5
6    VLAN0006                      active    Po6, Gig1/0/6
10   VLAN0010                      active
Vlan1                  [up/up]
    FE80::2E0:B0FF:FE91:84E
    3333:3333:3333:1111::1
Vlan2                  [up/up]
    FE80::2E0:B0FF:FE91:801
    3333:3333:3333:2222::1
Vlan3                  [up/up]
    FE80::2E0:B0FF:FE91:802
    3333:3333:3333:3333::1
Vlan4                  [up/up]
    FE80::2E0:B0FF:FE91:803
    3333:3333:3333:4444::1
Vlan5                  [up/up]
    FE80::2E0:B0FF:FE91:804
    3333:3333:3333:5555::1
Vlan6                  [up/up]
    FE80::2E0:B0FF:FE91:805
    3333:3333:3333:6666::1
Vlan10                 [up/up]
    FE80::2E0:B0FF:FE91:806
    1000::3
```